

NISQ-Aware Quantum Kernel SVM for Network Intrusion Detection: A Regime-Specific Benchmark on NSL-KDD

Minh Tuan Pham, Phuc Hao Do, Nguyen Nang Hung Van, Quang Anh Nguyen, and Quan Tran Anh Vo

Abstract—Quantum support vector machines (QSVM) built on the ZZ-feature map are routinely reported to help on network intrusion detection (NIDS), yet the conditions under which they help are rarely isolated. This paper asks where a NISQ-feasible quantum kernel is worth its cost, and answers with a protocol in which one factor is varied at a time: the embedding dimension is fixed by a lexicographic rule that charges explicitly for two-qubit gate count; the entangling layer is ablated against an otherwise identical control; the test distribution is shifted; and the training-set size is swept across two orders of magnitude under both the natural class prior and a rare-enriched one. Every comparison is paired within run over ten runs, tuned symmetrically for quantum and classical models alike, and corrected for multiplicity within baseline families. Against six baselines that now include a random forest and XGBoost, we find no general advantage: of 110 controlled comparisons, 21 favour the quantum kernel, 21 favour a classical baseline, and 68 are inconclusive. What we do find is structure. Under the natural class prior the ordering against every strong baseline reverses between $N = 2000$ and $N = 5000$, and the reversal survives freezing the hyper-parameters rather than re-tuning them; enriching rare attacks twelvefold, as the submitted protocol did, removes it. The selection rule transfers to UNSW-NB15 unchanged and returns six qubits rather than four, while the sample-complexity reversal does not transfer. Finally, spending a larger feature budget forces a wider circuit, and widening destroys the kernel: at eight qubits all 48 paired comparisons favour the classical model, with a mechanism visible directly in the Gram matrix, whose off-diagonal dispersion decays roughly twice as fast for the ZZ map as for its entanglement-free control and predicts macro- F_1 across widths. We report the result as a regime map, including the regimes in which the quantum kernel loses.

Index Terms—Quantum machine learning, quantum kernel, support vector machine, network intrusion detection, NSL-KDD, UNSW-NB15, NISQ, zzFeatureMap, kernel concentration, sample complexity, class-prior shift.

I. INTRODUCTION

Network intrusion detection is an attractive target for quantum machine learning. The data are tabular and moderate-dimensional, the decision is a classification, and the opera-

tional value of a small accuracy gain is high. Quantum kernel methods are the natural instrument: a feature map prepares a state whose fidelities form a positive semi-definite kernel that can be dropped into a standard support vector machine [11], [12], so the quantum component is a single well-isolated block inside an otherwise classical pipeline.

The literature reflects that attractiveness. Studies applying quantum kernels and variational classifiers to NSL-KDD [4] and its successors [25] report accuracies competitive with, and sometimes above, classical baselines. What they rarely report is the shape of the claim: at which training set size, against which class prior, at which circuit width, and against which classical opponent. Those choices are not incidental. We show below that varying only the training-set size, with everything else fixed, is enough to reverse the ordering between a quantum kernel and every strong classical baseline we tested. A benchmark conducted at one operating point can therefore report an advantage or a deficit and be internally consistent in either case.

This paper asks a narrower question than the literature usually asks: *where* is a NISQ-feasible quantum kernel worth its cost? The question is posed as a regime-identification problem (Problem 1) whose answers include the regimes where the answer is “nowhere we can detect”.

Our protocol varies one factor at a time. The embedding dimension is fixed by a lexicographic rule that charges explicitly for two-qubit gates, before any downstream metric is inspected. The entangling layer is ablated against an otherwise identical control. The test distribution is shifted four ways. The training-set size is swept over two orders of magnitude under both the natural class prior and a rare-enriched one. Every comparison is paired within run over ten runs, both quantum and classical models are tuned by the same procedure on the same held-out set, and p -values are Holm-corrected within baseline families. The baselines include a random forest and XGBoost, which the submitted version of this work did not.

The headline is a map, not a number. Of 110 controlled comparisons, 21 favour the quantum kernel, 21 favour a classical baseline, and 68 are inconclusive. Inside that map there is structure worth reporting:

- **An ordering that reverses with sample size.** Under the natural class prior, QSVM-ZZ is the second-worst model at $N = 100$ and the best at $N = 10^4$; the paired difference against XGBoost, the random forest and SVM-RBF changes sign between $N = 2000$ and $N = 5000$ in all six baseline-tuning-arm combinations, and survives

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freezing the hyper-parameters rather than re-tuning them. Enriching rare attacks twelvefold — the sampling our own submitted protocol used — removes the effect.

- **A selection rule that transfers.** The three-stage rule returns $n^* = 4$ on NSL-KDD and, applied without changing a threshold, $n^* = 6$ on UNSW-NB15, recovered on 10/10 independent subsets. The entanglement ablation transfers as well; the sample-size reversal does not.
- **A measured boundary.** Spending a larger feature budget forces a wider circuit, and widening destroys the kernel: at $K = 80$, where the rule returns $n^* = 8$, all 48 paired comparisons favour the classical model and the entanglement ablation reverses sign. The mechanism is visible in the Gram matrix, whose off-diagonal dispersion decays about twice as fast for the ZZ map as for its entanglement-free control — the ratio the $\binom{n}{2}$ versus n phase-term counting predicts — and correlates with macro- F_1 at $r = +0.77$.

We also report where the method loses. It degrades faster than either tree ensemble under class-prior shift; it is the least robust of all seven models to feature perturbation; and although it detects rare attacks far better than the tree ensembles, an RBF-kernel SVM detects them better still.

Relation to the submitted version

This paper is a substantially revised version of an earlier submission, and five of that version's claims do not survive the revised protocol. We state them here rather than in a footnote. (i) The claimed advantage in the low-data regime is reversed in direction. (ii) A reported rare-attack gain of 6.7 points with $d = +0.68$ is withdrawn: no rare-class metric was computed by the code that produced it. (iii) The former Theorem 1 was false, as one reviewer identified; it and the objective it maximised have been removed rather than repaired. (iv) The former Pareto stage filtered no candidate on this data and has been replaced. (v) With a tree ensemble in the comparison, QSVM-ZZ is no longer the top-ranked model at the main operating point. Sec. III-H gives the technical detail, including one error no reviewer raised.

II. BACKGROUND AND RELATED WORK

A. Quantum kernels

A quantum kernel embeds x into a state $|\phi(x)\rangle$ prepared by a parameterised circuit and reads off the fidelity as a similarity [11], [12]. Two results bound expectations. Alignment-based bounds [23], [27] say that a kernel better aligned with the labels admits a tighter generalisation guarantee at equal empirical risk, but say nothing about whether a given circuit produces such a kernel. In the other direction, fidelity kernels from expressive maps concentrate exponentially toward a constant as the qubit count grows [22], so the same expressivity that makes a map interesting makes its Gram matrix uninformative at width. Sec. V-G measures both effects here: the alignment gain is large, and the concentration is what eventually destroys it.

B. Quantum machine learning for intrusion detection

Applications of quantum kernels and variational classifiers to network intrusion detection have grown quickly, most using NSL-KDD [4] or a successor corpus [25]; a 2026 survey [40] covers the field through January of that year and reaches two conclusions that bear directly on this paper: most published approaches rely on simulated quantum environments and legacy datasets, and evaluation practice is inconsistent across studies. The recurring pattern is a single operating point — one qubit count, one training-set size, one class prior — with an accuracy compared against a small set of classical baselines.

The most directly comparable study is the benchmark of Cirillo *et al.* [36], which evaluates Pegasos-QSVC, a variational classifier and a hybrid quantum–classical network on ToN_IoT and NSL-KDD under ideal and three backend-derived noise models, and reports Pegasos-QSVC best at 94.60% accuracy and 94.13% F_1 on NSL-KDD.

Two of their findings bear on ours. They compare against SVM, XGBoost, random forest and a neural network at matched dimensionality and conclude that the classical models retain higher peak accuracy — XGBoost reaching 99.1% on ToN_IoT — while the quantum models stay competitive in low-dimensional feature spaces; our own ranking agrees (Sec. V-B). And they sweep the qubit count from 2 to 10 against two-qubit gate count, finding performance concentrated in a 20–35 CNOT band and degrading beyond about 60. Our operating point of 24 CNOTs sits inside that band, and the collapse we report at 112 is consistent with their threshold.

The two studies diverge on the mechanism, and that divergence is part of our contribution: they attribute the degradation to two-qubit gate error accumulating on noisy backends, whereas we observe the same collapse under *exact, noiseless* simulation and trace it to concentration of the Gram matrix. Hardware noise is not a necessary ingredient; the bound is intrinsic to the kernel.

What that study does not vary is the training-set size, fixed at a single 15% class-balanced subsample (15,114 training instances for NSL-KDD) drawn under a random 80/20 split rather than the official partitions.

C. Benchmarking studies of quantum models

Two broad benchmarks are relevant precisely because they are sceptical. Bowles, Ahmed and Schuld [37] evaluate twelve QML models over 160 datasets — predominantly synthetic, so that data properties can be controlled — in noise-free statevector simulation at up to 18 qubits, against an MLP, an RBF-kernel SVM and a CNN, and find that quantum models rarely beat well-tuned classical baselines. Schnabel and Roth [38] run over 20,000 tuned models comparing fidelity and projected quantum kernels across five dataset families, at up to 15 qubits against RBF-kernel baselines, and use correlation analysis to identify which design choices matter. Both conclude that general-purpose feature maps seldom win once the classical comparison is done properly, and our results agree with them at the aggregate level.

Neither varies the training-set size: Bowles *et al.* subsample fixed counts per task, and Schnabel and Roth fix $M = 240$ training and $M' = 60$ test points throughout.

Concurrently with this revision, Gillani *et al.* [39] benchmark quantum and classical models for intrusion detection across four corpora, including both of ours, at eight qubits with a 4/6/8/12-qubit ablation, against five tuned classical baselines including a random forest and XGBoost, with Benjamini–Hochberg control over 108 comparisons on NSL-KDD. Their conclusion — that reported quantum gains are largely attributable to the classical preprocessing front-end once the same front-end is given to classical models — corroborates, from a different direction, the decomposition we report in Sec. V-D. We do not claim breadth over that study: they evaluate four datasets to our two, at eight qubits to our four and six, with a stricter error-rate criterion. What they hold fixed is the training-set size — the full 125,973-record training partition on NSL-KDD — and that is where our contribution lies.

D. Bounded advantage windows in adjacent domains

The question this paper asks is being asked elsewhere, and in at least one adjacent domain it has been given the same shape. Carducci [41] opens by arguing that malware analysts “do not need another general claim that quantum computing is either broadly superior or broadly impractical” but rather “need to know which kinds of malware representations justify quantum effort” — which is, on a different task, the reframing this revision makes.

That study places malware representations on an eleven-rank representational-complexity scale and reports that the measured advantage is confined to an *intermediate* window: at the co-occurrence and local-order representations the quantum workflow retrieves malware where the classical baseline retrieves none (4.29% against 0%), while classical methods remain stronger both at the lower-complexity statistical representations and at the higher-complexity ones. The advantage survives parameter tuning, which the author reports specifically to rule out a favourable-setting artefact, and the stated practical implication is to prefer classical methods wherever mature inductive biases already exist.

Three features of that result recur in ours, on a different task and with a different quantum primitive: the advantage is bounded on *both* sides rather than growing with capability; the authors test explicitly whether re-tuning explains it, as our frozen and tuned arms do; and the region where the quantum method helps is characterised by *local pairwise structure*, which is what the $\binom{n}{2}$ pairwise phase terms of the ZZ map encode and what Sec. V-G shows to stop paying off once n grows.

We could not obtain that full text through our institutional access, so the comparison rests on its abstract, introduction and index terms; we could not establish its qubit count or whether it varies training-set size. We present the parallel as suggestive rather than as evidence, and do not tabulate the study in Table I, whose cells are read from full texts.

E. Positioning

Table I places this work against those four studies on the axes that distinguish them; each cell was read from the full text

rather than from an abstract. The distinguishing axis is not the model, the dataset, the circuit width or the noise treatment, on all of which we are comparable or narrower — two of the four vary the qubit count, and two compare against tree ensembles. It is that *none of the four sweeps the training-set size*, which each fixes at a single value, and the training-set size is the variable that decides the answer here: the ordering between QSVM-ZZ and every strong classical baseline reverses inside the range $N \in [10^2, 10^4]$ under the natural class prior. A benchmark conducted at a single N — as all four are, and as our own submitted version was — can therefore report either outcome and be internally consistent.

We take the reviewers’ point that benchmarking alone is not a contribution, and we note that the two studies most often cited as establishing what is already known [37], [38] are themselves benchmarks without a new kernel, feature map, algorithm, hardware demonstration or theoretical result. What makes a benchmark worth publishing is not its novelty of apparatus but whether it changes what the field believes. We report an ordering that reverses with sample size, a selection rule whose output transfers across datasets, and a measured mechanism — Gram-matrix concentration — that predicts where the method stops working. The last of these is the closest we come to a theoretical contribution, and we present it as an empirical regularity rather than as a theorem, which is what our evidence supports.

III. NISQ-AWARE QSVM FRAMEWORK

A. Feature map, kernel and cost

The ZZFeatureMap on n qubits with r repetitions prepares

$$|\phi(\mathbf{x})\rangle = [U_{\Phi}(\mathbf{x}) H^{\otimes n}]^r |0\rangle^{\otimes n}, \quad (1)$$

where H is the Hadamard gate and U_{Φ} is diagonal in the computational basis,

$$U_{\Phi}(\mathbf{x}) = \exp\left(i \sum_{i=1}^n x_i Z_i + i \sum_{1 \leq i < j \leq n} \frac{1}{2} \varphi_{ij}(\mathbf{x}) Z_i Z_j\right), \quad (2)$$

with the pairwise phase $\varphi_{ij}(\mathbf{x}) = 2(\pi - x_i)(\pi - x_j)$. The induced kernel is the squared fidelity

$$K^Q(\mathbf{x}, \mathbf{x}') = |\langle \phi(\mathbf{x}') | \phi(\mathbf{x}) \rangle|^2, \quad (3)$$

and the circuit cost we charge for it is

$$Q(n) = \frac{Q_{\text{raw}}(n)}{Q_{\text{raw}}(n_{\text{max}})}, \quad Q_{\text{raw}}(n) = 10n^2 - 8n. \quad (4)$$

Removing the pairwise term of Eq. (2) while leaving everything else unchanged gives the ZFeatureMap, which is the control used throughout the entanglement ablation.

B. Problem statement

Let $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^N$ be a labelled sample of network records with $\mathbf{x}_i \in \mathbb{R}^d$ and y_i one of five classes, and let K_n^Q denote the kernel of Eq. (3) on an n -qubit map. A quantum kernel is not free: evaluating K_n^Q over a training set of size N requires $\Theta(N^2)$ circuit evaluations, each costing $Q(n)$ two-qubit gates on hardware whose error rate is dominated

TABLE I

POSITIONING AGAINST THE CLOSEST BENCHMARKING STUDIES. COLUMNS ARE THE WORKS THE REVIEWERS POINTED US TO, PLUS THE CLOSEST CONCURRENT QML-IDS BENCHMARK. EVERY CELL WAS READ FROM THE FULL TEXT OF THE CITED WORK, EXCEPT CARDUCCI (2026), WHICH WE COULD NOT OBTAIN AND WHICH IS THEREFORE DISCUSSED IN THE TEXT BUT NOT TABULATED; “N/R” MEANS THE ITEM IS NOT REPORTED BY THAT WORK AT ALL. THE LAST THREE ROWS ARE THE AXES ON WHICH THIS WORK IS NOT DUPLICATED; THE ROWS ABOVE THEM ARE WHERE WE ARE COMPARABLE OR WEAKER.

	Bowles et al. [37]	Schnabel Roth [38]	&	Cirillo et al. [36]	Gillani et al. [39]	This work
Domain Datasets	general QML 6 tasks / 160 sets	general QKM 5 families / 64 sets		network IDS NSL-KDD, ToN_IoT	network IDS NSL-KDD, UNSW-NB15, CICIDS2017, NF- ToN-IoT-v2	network IDS NSL-KDD, UNSW- NB15
Test split	synthetic splits	fixed 240/60		random 80/20 on a 15% class-balanced subsample	official splits	official splits
Quantum model	12 QML models	FQK, PQK		Pegasos-QSVC, VQC, HQNN	hybrid VQC, IQP fi- delity & projected kernels	ZZ / Z fidelity kernels
Qubits	up to 18	up to 15 (= fea- tures)		2–10 swept, best 7	8 (abl. 4/6/12)	4 (NSL), 6 (UNSW), <i>selected by rule</i>
RF + XGBoost baselines	no	no		yes	yes	yes
Repeats	n/r	>20,000 models (HPO)		4 seeds	3–5 seeds	10 runs
Multiplicity control	n/r	correlation analy- sis		none (mean $\pm$ sd)	BH-FDR over 108 comparisons	Holm within baseline family
Noise model	noise-free	noise-free		ideal + noisy sim- ulator	NISQ noise sweep	ideal, finite-shot, noisy simulator
Training-set size swept	no	no		no	no (width ablation only)	yes , $N=10^2\text{--}10^4$
Regime decomposition	aggregate ranking	aggregate correla- tions		F_1 vs. CNOT count	per-dataset verdict	110 verdicts over prior, composition, shift, size
Selection rule transferred	—	—		—	—	yes : $n^*=4 \rightarrow 6$ un- changed

by exactly those gates. The question this paper answers is therefore not whether K_n^Q can be made to win somewhere, but where it is worth this cost.

Assumption 1 (NISQ cost model). *Circuit cost is charged by width alone, through the CNOT-weighted count of Eq. (4), normalised by its value at the largest candidate so that $Q(n) \in (0, 1]$. The quantity it tracks is the entangling-gate budget: an r -repetition $ZZ\text{FeatureMap}$ with full entanglement places $2r\binom{n}{2}$ two-qubit gates, which at the operating point $n = 4$, $r = 2$ is 24 and at $n = 8$ is 112. The $\Theta(N^2)$ kernel evaluations are excluded from Q because they scale with N rather than n and so do not affect the choice of width; they are, however, the dominant cost of deploying the method at all, which is why Sec. VI reports the N at which the kernel becomes competitive and not merely that it does.*

Assumption 1 is what makes the selection rule of Definition 1 a statement about hardware rather than about taste, and it is also its main limitation: a device whose dominant error source is idle-time decoherence rather than entangling-gate infidelity would call for a different Q .

Problem 1 (Regime identification). *Given a family of operating conditions — training-set size, class prior, attack composition, test distribution, and embedding width — determine, for each condition, whether $K_{n^*}^Q$ is preferable to a classical baseline, at a stated level of statistical confidence, and report*

the conditions in which it is not.

Problem 1 is deliberately weaker than the claim the submitted version made. It admits negative answers as results, and most of the answers we obtain are negative or inconclusive.

C. Pipeline

The path from raw records to a decision is: one-hot encoding to $d = 122$ dimensions; an ANOVA F -test retaining K features; PCA to n components; a min-max scaler mapping each component into $[0, \pi]$, the domain on which the feature map’s phases are defined; the $ZZ\text{FeatureMap}$ of Eq. (1) on the resulting $\mathbf{z} \in [0, \pi]^n$; and a soft-margin SVC on the precomputed Gram matrix.

Two properties of this pipeline carry the experimental design.

One component varies at a time. The entanglement-free control differs from the main path in exactly one block: the $Z\text{FeatureMap}$ applies the same Hadamard and single-qubit phase layers and omits the pairwise term of Eq. (2). It has the same qubit count, the same repetitions, the same classical front-end, and the same classifier, so a difference between the two branches is attributable to entanglement and to nothing else. The same discipline governs the other three contributions: C1 fixes the representation before any downstream metric is inspected, C3 varies only the test distribution, and C4 varies only N .

The front-end is refit wherever the classifier is refit. In the primary arm, feature selection, projection and scaling are re-estimated at every training-set size on that size's training fold alone. This costs comparability across sizes — the representation at $N = 100$ is not the representation at $N = 10^4$ — and buys the guarantee that no representation is ever informed by more data than the classifier it feeds. We regard that trade as mandatory in a study whose headline concerns how behaviour changes with N : freezing a front-end fit on the largest subset would leak precisely the quantity under investigation. Sec. V-D reports both arms.

D. Kernel evaluation

The kernel of Eq. (3) is evaluated exactly rather than by sampling. Because the ZZFeatureMap is diagonal in the computational basis after each Hadamard layer, $|\phi(\mathbf{x})\rangle$ has a closed form, and the Gram matrix follows from dense linear algebra. Finite-shot estimation and backend-derived noise are then applied as *separate* conditions (Sec. V-B) rather than being folded into the main results, which keeps sampling error from being confused with the effects under study.

E. Preliminaries

We use three standard facts, stated here for reference rather than proved, since each is established in the literature.

- (F1) *Positive semi-definiteness.* For any finite sample, the Gram matrix $[K^Q(\mathbf{x}_i, \mathbf{x}_j)]_{i,j=1}^N$ induced by a feature map is symmetric and positive semi-definite, being a Hadamard product of two Hermitian PSD factors, by the Schur product theorem [17]. This is what licenses the use of K^Q inside the dual soft-margin SVM [16], [17], with the quantum kernel slotting in exactly as in [11], [12].
- (F2) *Kernel–target alignment and generalisation.* Higher alignment together with comparable empirical risk implies a tighter generalisation bound [23], [27]; we use the bound exactly as stated in [23] and derive nothing new from it. Its role here is diagnostic: two kernels with the same empirical risk may differ in the bound because the hypothesis classes their geometries induce differ.
- (F3) *Concentration at large width.* Fidelity kernels built from expressive feature maps concentrate toward a constant as the number of qubits grows [22], which is an independent reason — beyond gate count — to prefer a small embedding dimension.

Remark 1. In the submitted version these three facts appeared as Propositions 1, 3 and 4 with accompanying proofs. We agree with the reviewer that presenting established results in that form was inappropriate, and that the proof of the former Proposition 3 restated [23] rather than deriving anything. They are now cited as background.

Lemma 1 (Local expansion of the ZZ kernel at $r = 1$). *Let $r = 1$ and let $\mathbf{x}, \mathbf{x}' \in [0, \pi]^n$ with $\mathbf{x}'$ sufficiently close to $\mathbf{x}$.*

Then the kernel of Eq. (3) admits the second-order expansion

$$K^Q(\mathbf{x}, \mathbf{x}') = 1 - \sum_{i=1}^n \Delta_i^2 - \frac{1}{4} \sum_{1 \leq i < j \leq n} \Delta \varphi_{ij}^2 + O(\|\mathbf{x} - \mathbf{x}'\|^4), \quad (5)$$

where $\Delta_i := x_i - x'_i$ and $\Delta \varphi_{ij} := \varphi_{ij}(\mathbf{x}) - \varphi_{ij}(\mathbf{x}')$.

Remark 2 (The expansion does not extend to $r \geq 2$). The restriction to one repetition is not cosmetic. For $r \geq 2$ the interleaved Hadamard layers make $U_\Phi(\mathbf{x}')^\dagger U_\Phi(\mathbf{x})$ non-diagonal and no pair of coefficients renders the two-invariant form correct: a least-squares fit at $r = 2$ attains only $R^2 = 0.60$ – 0.94 , with coefficients that change sign between expansion points. The quantity $1 - K^Q$ stays a quadratic form in $\mathbf{x} - \mathbf{x}'$ at every r (fitting to $R^2 > 0.999$) but is no longer expressible in these two invariants. Since our operating point is $r = 2$, we use Lemma 1 for the structural reading below and not as a numerical statement. See `runners/verify_lemma1.py`.

Lemma 1 is retained because, unlike (F1)–(F3), it is specific to the comparison this paper makes: the linear part $\sum_i \Delta_i^2$ is the quantity a Gaussian RBF kernel controls through its bandwidth, and the pairwise part expands into the monomials a degree-2 polynomial kernel can represent. What the classical kernels lack is the higher-order oscillation carried by the exponentials, bounded above by 1 because K^Q is a squared fidelity whereas the polynomial inner product grows unboundedly with $\|\mathbf{x}\|$. This is why SVM-POLY2 is the sharpest classical comparator, and Sec. V-B confirms that it is. The proof is given in Appendix A.

F. C1: A Three-Stage Rule for the Embedding Dimension

The submitted version selected n by Pareto filtering followed by a weighted scalarisation $J(n) = \alpha V(n) + \beta \tilde{F}(n) - \gamma Q(n)$. That construction has been removed; Sec. III-H states why. It is replaced by a rule that is *lexicographic* rather than compensatory, so no weights are involved and no trade-off among the three objectives has to be asserted.

Definition 1 (Three-stage selection). *Let $\mathcal{N}$ be the candidate set, $V(n)$ the cumulative explained variance, $\text{KTA}(n)$ the kernel–target alignment of the induced Gram matrix, and $Q(n)$ the CNOT-weighted cost of Eq. (4). Fix $T = 0.85$ and $\epsilon = 0.05$ before inspecting any downstream metric. Then*

$$\mathcal{N}_1 := \{n \in \mathcal{N} : V(n) \geq T\}, \quad (6)$$

$$\mathcal{N}_2 := \left\{n \in \mathcal{N}_1 : \text{KTA}(n) \geq (1 - \epsilon) \max_{m \in \mathcal{N}_1} \text{KTA}(m)\right\}, \quad (7)$$

$$n^* := \arg \min_{n \in \mathcal{N}_2} Q(n). \quad (8)$$

The three stages ask three different questions and are therefore not interchangeable. Stage 1 asks whether the representation still carries the signal; stage 2 asks whether the kernel on that representation is aligned with the labels; stage 3 breaks the remaining tie by hardware cost. Stage 1 is what removes $n = 2$ on NSL-KDD despite its being the most aligned candidate overall: a kernel can be well aligned on a representation that has already discarded a quarter of the variance.

Three-stage dimension selection reproduces on an unseen dataset

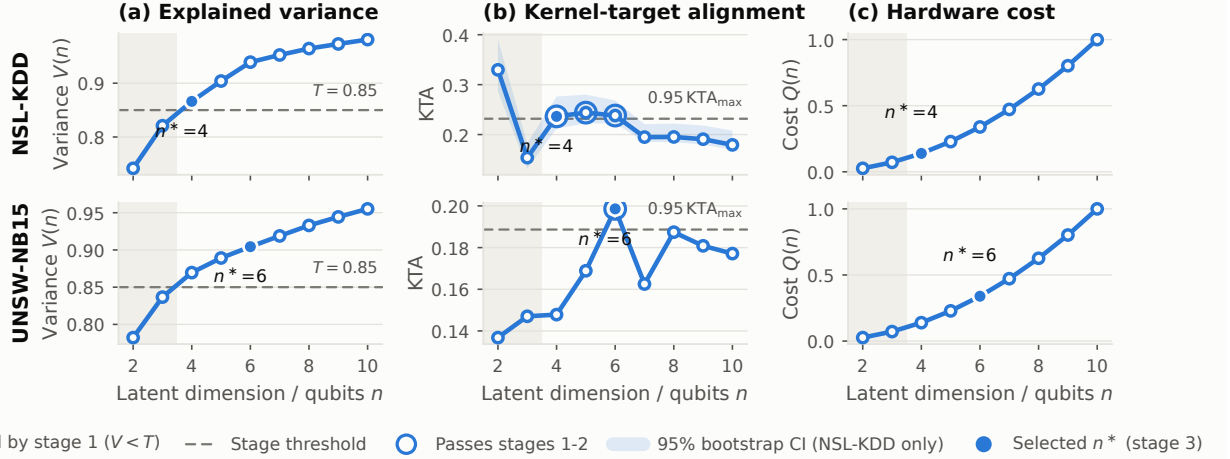


Fig. 1. **The three-stage dimension-selection rule, applied unchanged to two datasets.** Rows: NSL-KDD (top), UNSW-NB15 (bottom). Columns: (a) cumulative explained variance $V(n)$, (b) kernel-target alignment, (c) CNOT-weighted cost $Q(n)$. Stage 1 discards $V(n) < 0.85$ (shaded), stage 2 keeps $KTA \geq 0.95 KTA_{\max}$ (open circles), stage 3 takes the cheapest survivor (filled). The same rule returns $n^* = 4$ and $n^* = 6$.

On NSL-KDD the rule gives $\mathcal{N}_1 = \{4, \dots, 10\}$, a maximum alignment of 0.2439 at $n = 5$, hence a stage-2 threshold of 0.2317, $\mathcal{N}_2 = \{4, 5, 6\}$ and $n^* = 4$. Applied unchanged to UNSW-NB15 it gives $\mathcal{N}_2 = \{6\}$ and $n^* = 6$, recovered on 10/10 independent subsets and under every threshold perturbation we tested (Fig. 1). We therefore present C1 as a transferable *procedure* whose output is a property of the dataset, and not as the constant $n = 4$.

G. Identifying the Entanglement Contribution

The submitted version certified the entanglement effect with a two-sample z -test over $m = 5$ subsets (former Proposition 4). Two things are wrong with that. The test treats the two arms as independent when they in fact share the training subset, discarding the pairing that makes the comparison informative; and five subsets is too few for the normal approximation it invokes.

The revision compares the arms *pairwise within a run*. For each of ten runs we compute $\Delta = KTA_{ZZ} - KTA_Z$ on the same subset and report the mean with a t -based 95% interval, a Wilcoxon signed-rank test, and the paired effect size d_z . Because each contribution tests several baselines, p -values are Holm-corrected within baseline families (*entanglement*, *classical kernel*, *strong tabular*), and every verdict reported in this paper is the corrected one. This is strictly more conservative than the former z -test and assumes nothing beyond symmetry of the paired differences.

H. Corrections to the Submitted Version

Four defects in the submitted theory section are corrected here; two were identified by reviewers and two we found ourselves, and we report them on the same footing. The accompanying letter gives the full derivations.

(i) **Theorem 1 was false.** Its proof asserted $\tilde{F}(4) > \tilde{F}(3)$, whereas Table III of the same submission reports $0.471 < 0.628$. With the correct values the objective is maximised at $n = 2$ ($J = 0.551$), not at $n = 4$. We thank the reviewer for identifying this.

(ii) **The Pareto stage filtered nothing.** $V(n)$ increases monotonically while $\tilde{F}(n)$ decreases and $Q(n)$ increases, so every candidate is Pareto-optimal. Theorem 1 and J are removed rather than repaired, because this second defect shows the construction was not doing the work attributed to it.

(iii) **The former Proposition 2 had incorrect coefficients.** It asserted the expansion with r^2 on both sums. The correct weights are 1 and $\frac{1}{4}$ (Lemma 1), and the r^2 prefactor does not hold at all for $r \geq 2$ (Remark 2). The difference is not marginal: with the submitted coefficients the residual decays as $\|\cdot\|^2$ rather than $\|\cdot\|^4$, so the claimed second-order accuracy is not attained. No reviewer raised this; we found it while transcribing the proof. The consequence drawn from it — that SVM-POLY2 is the sharpest classical comparator — is unaffected, depending on which monomials appear and not on their coefficients.

(iv) **Table III mislabelled its third column.** The caption defined $\tilde{F}(n) = 1/\text{DBI}(n)$, but the tabulated values are the ANOVA F statistic. No conclusion depends on it, the quantity having been dropped along with J , but we record it for completeness.

IV. EXPERIMENTAL SETUP

A. Data and preprocessing

NSL-KDD [4] is used with its official partitions. The training partition holds 125,973 records and the test partition 22,544; the class mix is Normal 53.5%, DoS 36.5%, Probe 9.3%, R2L 0.8% and U2R 0.04%, so the two rare classes together account for 0.83% of the data. One-hot encoding of

the three categorical fields expands the feature space from 41 to 122 dimensions. UNSW-NB15 [25] is used with its official CSV partitions and the same pipeline, expanding to 189 columns.

Two properties of these corpora are relevant to how the numbers should be read and are quantified in Sec. VII: NSL-KDD has 610 test rows identical to a training row in both features and label, and UNSW-NB15 is far more heavily duplicated.

B. Protocol

Repetitions. Every reported quantity is computed over ten independent runs with fixed seeds $\{100, \dots, 109\}$. Subsets are drawn stratified on the four-way label $\{\text{Normal}, \text{DoS}, \text{Probe}, \text{R2L} \cup \text{U2R}\}$, which guarantees at least one rare-attack instance even at $N = 100$. The submitted version used five seeds; the increase to ten was prompted by the observation that five is a thin base for the interval estimates we report.

Two test sets. Results are reported both on the fixed 300-sample stratified split used by the submitted version, which keeps continuity with it, and on the complete KDDTest⁺ partition. The latter is the operative one for any deployment reading, and the two differ enough that quoting only the former would overstate absolute performance.

Symmetric tuning. The submitted version fixed the quantum regulariser at $C = 1.0$ while selecting C for the classical SVMs on a validation fold, and described this as avoiding bias. It does the opposite: it grants one family a search the other is denied. In this revision every model, quantum and classical, is tuned by the same procedure on the same disjoint tuning set of 2,000 records, and results are reported for two arms — *tuned per N* , in which the search is repeated at each training-set size, and *frozen*, in which hyper-parameters are held at the values selected once. Because the quantum kernel does not depend on C , its grid costs only repeated fits on an already-computed Gram matrix.

Zero leakage. All transformers are fit on the training fold alone and applied unchanged to the test fold. In the primary arm the entire front-end, including feature selection and the projection, is refit at each N , so a representation never sees more data than the classifier it feeds. The nested subsets satisfy $\mathcal{D}_{N_1} \subset \mathcal{D}_{N_2} \subset \dots$, so variation across sizes reflects added data rather than a changed sample.

C. Statistical methodology

Every comparison is paired within run: for run i we form $\Delta_i = m(\text{QSVM-ZZ}) - m(\text{baseline})$ on the same subset and the same test rows. We report the mean of Δ with a t -based 95% confidence interval, a Wilcoxon signed-rank test, the paired effect size d_z , and the fraction of runs with $\Delta > 0$.

Because each contribution tests several baselines, p -values are Holm-corrected within baseline families — *entanglement* $\{\text{QSVM-Z}\}$, *classical kernel* $\{\text{linear}, \text{poly2}, \text{RBF}\}$, and *strong tabular* $\{\text{random forest}, \text{XGBoost}\}$ — and every verdict in this paper is the corrected one. A comparison is called favourable to one side only when the corrected $p < 0.05$ and the

confidence interval excludes zero; otherwise it is inconclusive, and we report it as such rather than as a win.

D. Implementation and reproducibility

Experiments use NumPy 2.4, SciPy 1.17, scikit-learn 1.8, XGBoost 3.3, Qiskit 2.3 with qiskit-machine-learning 0.9, and Qiskit Aer 0.17. Quantum kernels are evaluated as exact statevector fidelities. Because the `zzFeatureMap` is diagonal after each Hadamard layer, the state can be written in closed form rather than simulated gate by gate, which we verified against the Qiskit reference to within 4×10^{-15} and which makes the sweeps in Sec. V tractable.

Two sources of run-to-run variation deserve mention because they set the precision of what follows. XGBoost with the histogram tree method varies by about ± 0.001 macro- F_1 between machines even with a single thread, so we report its values to three decimals and never rest a claim on a smaller difference. The fidelity statevector kernel exposes no seed of its own; its determinism is controlled through the global Qiskit seed, which we pin.

Reproducibility. The complete implementation, the frozen protocol files, every intermediate artifact, all figure-generation scripts, and four audit scripts are released at https://github.com/anhquan1111/QSVM_NSLKDD, at commit 19beb18.¹

The audits are not decorative, and running them is the intended way to check this paper. `audit_c4.py` (100 checks) recomputes every published statistic from the raw per-run records using an independent implementation rather than calling the pipeline’s own statistics functions, since using the code that produced a number to verify that number lets shared errors through. `audit_figures.py` (36) checks that each figure is newer than its inputs and that every plotted number is re-derived from the artifacts. `audit_prose.py` (115) re-derives every number written in the text of this paper. `verify_lemmal.py` (15) checks the expansion of Lemma 1 against the exact kernel. Together they found four errors in our own revision code before publication, one of which changed a reported n^* .

V. RESULTS

A. Choosing the embedding size, and what it costs

The feature budget K and the qubit count n are not independent, and the selection rule makes the coupling explicit.

Fig. 2(a) sweeps K with a classical proxy. Raw macro- F_1 rises monotonically to 0.9727 at the full $K = 122$, and after projection to four components it peaks at 0.9283 at $K = 80$; at the inherited operating point $K = 20$ the projected value is 0.9007. So $K = 20$ is not the accuracy optimum, and we do not present it as one.

What $K = 20$ is, is the largest budget the rule can serve with a four-qubit circuit. Applying the three-stage rule of Sec. III at $K = 20$: the variance gate $V(n) \geq 0.85$ admits $n \in \{4, \dots, 10\}$; the alignment gate $\text{KTA} \geq 0.95 \text{KTA}_{\max}$, with $\text{KTA}_{\max} = 0.2439$ at $n = 5$ and hence a threshold of 0.2317, narrows this to $\{4, 5, 6\}$; and minimising $Q(n)$ selects

¹Commit hash to be updated to the final revision commit before publication.

The accuracy a larger K buys is paid for in qubits

Five-fold stratified CV on the full training set; SelectKBest refit inside each fold. $K = 20$ (dashed) is the value inherited from the submitted protocol.

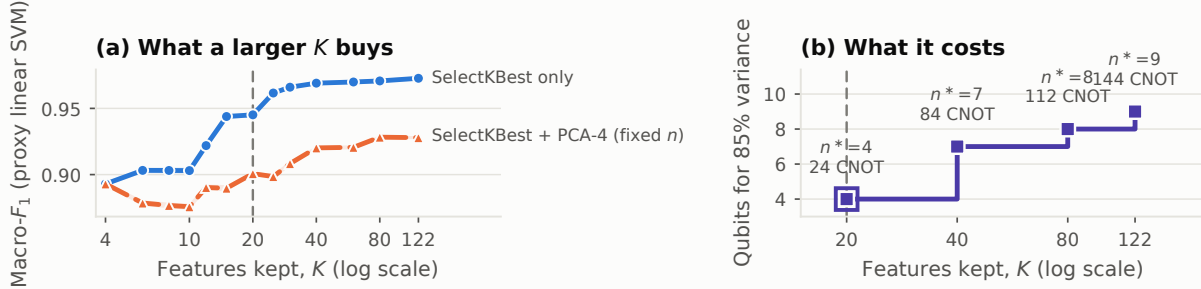


Fig. 2. **The accuracy a larger feature budget buys is paid for in qubits.** (a) Macro- F_1 of a proxy linear SVM against K , with and without the PCA stage, under five-fold stratified cross-validation; SelectKBest is refit inside each fold, so the curve carries no selection leakage. (b) The qubits the rule requires at that K , annotated with the resulting two-qubit gate count. $K = 20$ is not the accuracy optimum; it is where the circuit is still cheap.

$n^* = 4$, that is 24 two-qubit gates. Fig. 1 shows the three stages acting in sequence.

Re-running the same rule at larger budgets (Fig. 2(b)) gives $n^* = 4, 7, 8, 9$ for $K = 20, 40, 80, 122$. The mechanism is the variance gate: spreading the same variance over more retained features pushes $V(n)$ down at every n , so the gate opens later. This is the honest reading of $K = 20$. It is not an elbow in accuracy — accuracy keeps improving — it is the point at which the circuit is still cheap. Moving from $K = 20$ to $K = 80$ buys 0.028 macro- F_1 for the classical proxy and costs a rise from 24 to 112 two-qubit gates. Sec. V-G reports what happens when that price is actually paid.

B. Does entanglement help at the operating point?

At $n^* = 4$ we compare QSVM-ZZ against an otherwise identical circuit with the entangling layer removed, and against five classical baselines (Fig. 4). The answer splits by what is measured.

In kernel geometry the entangling layer matters, and by a wide margin: the paired alignment gain is $\Delta KTA = +0.1378$ [$+0.1267, +0.1489$], $d_z = 8.91$. In end-task accuracy it does not resolve: $\Delta F_1^{\text{macro}} = +0.0114$ [$-0.0054, +0.0281$], $p = 0.23$. The interval crosses zero and we report the comparison as inconclusive. Alignment is a property of the Gram matrix; whether it converts into accuracy depends on the classifier and the task, and here it does not measurably convert.

Across all seven models the ordering is XGBoost 0.8503 [$0.8412, 0.8595$], QSVM-zz 0.8469 [$0.8321, 0.8616$], random forest 0.8446, SVM-RBF 0.8362, QSVM-z 0.8355, poly-2 0.8327, linear 0.8137. The submitted version reported QSVM-ZZ at the top of this table because it did not include a tree ensemble; with one included, QSVM-ZZ is second and the intervals overlap. We therefore describe this as an ordering of point estimates and not as a separation.

Under the backend-derived noise model the conclusion is unchanged. QSVM-ZZ attains 0.8665 under exact statevector simulation, 0.8598 with finite shots, and 0.8728 under FakeManilaV2 noise, with QSVM-Z at 0.8498, 0.8464 and 0.8528; the noisy value is not lower than the ideal one, which

we read as noise being immaterial at this scale rather than as noise helping.

C. Distribution shift

Four shifts are applied to a fixed operating point (Fig. 5).

Class-prior shift. As the attack fraction is driven from its natural value to 30, 50 and 70%, QSVM-ZZ separates from the entanglement-free control ($+0.0237$ and $+0.0367$ at 50 and 70%, both corrected-significant) and from the linear kernel, but it degrades *faster* than either tree ensemble: across the shift it loses 0.032 macro- F_1 against 0.021 for XGBoost and 0.025 for the random forest, and at 70% attack the comparison against XGBoost is classical-favourable (-0.0242 , $p_{\text{Holm}} = 0.039$). The submitted version named prior shift as its strongest evidence of advantage. Against strong tabular baselines it is not; we correct that claim here.

Attack composition. On a 50/50 Normal-DoS mixture QSVM-ZZ is favourable against all four kernel baselines, with effect sizes from 1.56 to 3.02, and inconclusive against both tree ensembles ($+0.0014$ and $+0.0006$). The pattern is consistent: the quantum kernel competes with other kernels, and the tree ensembles are a separate matter.

Temporal shift. On the harder KDDTest-21 partition QSVM-ZZ degrades more than the control and more than the linear and poly-2 kernels (all three classical-favourable) and is inconclusive against SVM-RBF and both ensembles. No reading of this regime favours the quantum kernel.

Feature perturbation. Under additive noise of increasing σ , QSVM-ZZ's degradation slope is steeper than every baseline's — six of six comparisons classical-favourable, slopes from -0.69 to -1.11 , all effect sizes beyond 2.8. This is the clearest negative result in the paper and we present it as such.

D. Sample complexity: an ordering that reverses

The submitted version claimed an advantage in the low-data regime from a sweep that stopped at $N = 1000$. Extending it to $N = 10^4$ under the natural class prior reverses the claim and, in doing so, produces the paper's main new finding (Fig. 6).

Entanglement moves the kernel geometry far more than the score

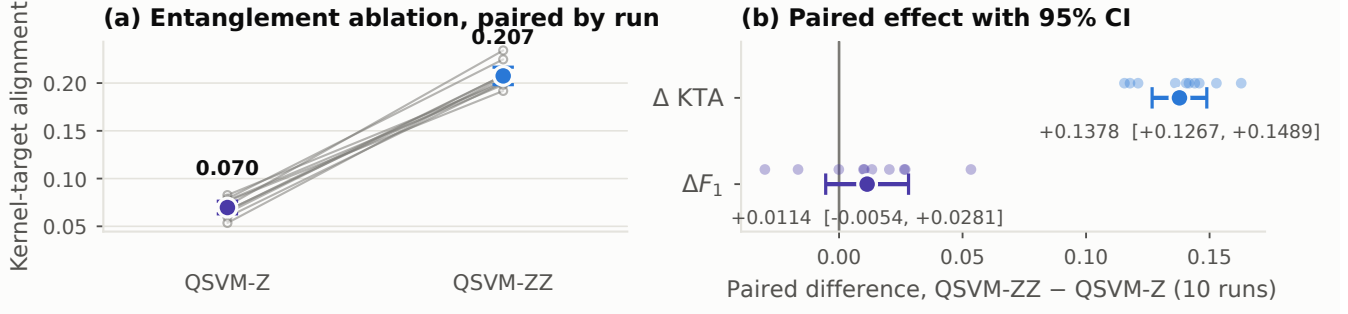


Fig. 3. **Entanglement moves the kernel geometry far more than it moves the score.** (a) Kernel–target alignment for the two feature maps, paired within each of ten runs. (b) The paired differences. Removing the entangling layer costs 0.1378 alignment [0.1267, 0.1489], an interval far from zero, but only 0.0114 macro- F_1 [-0.0054, 0.0281], an interval that contains it.

With strong tabular baselines the quantum kernel no longer leads

Dots are individual runs; markers are the mean with a 95% confidence interval. The intervals overlap: this is an ordering of point estimates, not a significant separation.

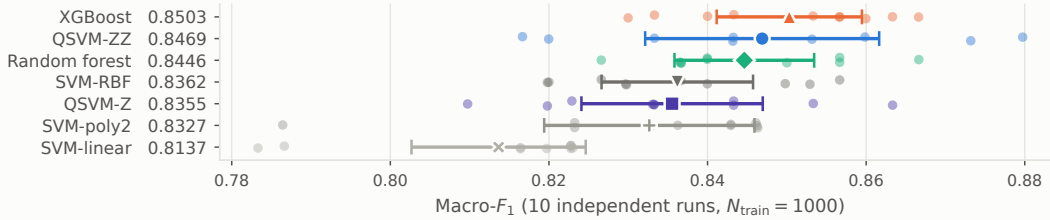


Fig. 4. **With strong tabular baselines the quantum kernel no longer leads.** Macro- F_1 over ten independent runs at $N_{\text{train}} = 1000$; dots are individual runs, markers are means with 95% confidence intervals. Adding a random forest and XGBoost under the identical protocol changes the ordering: XGBoost attains 0.8503 against QSVm-ZZ’s 0.8469, with overlapping intervals.

TABLE II

THE CROSSOVER IS NOT AN ARTEFACT OF PER- N RE-TUNING. PAIRED MEAN $\Delta F_1 = \text{QSVm-ZZ} - \text{BASELINE}$ OVER 10 RUNS ON THE FULL KDDTEST+ SET, NATURAL CLASS PRIOR. *frozen* KEEPS THE HYPER-PARAMETERS FIXED AT THEIR C2 VALUES; *tuned* RE-TUNES EVERY MODEL, QUANTUM AND CLASSICAL ALIKE, AT EACH N . ALL SIX BASELINE-ARM COMBINATIONS CHANGE SIGN BETWEEN $N = 2000$ AND $N = 5000$ (BOLD ROW).

N	XGBoost		Random forest		SVM-RBF	
	frozen	tuned	frozen	tuned	frozen	tuned
100	-0.0393	-0.0812	-0.0242	-0.0705	-0.0145	-0.0307
200	-0.0236	-0.0641	-0.0196	-0.0627	-0.0073	-0.0284
500	-0.0213	-0.0293	-0.0115	-0.0293	-0.0118	-0.0292
1000	-0.0309	-0.0289	-0.0171	-0.0196	-0.0021	-0.0028
2000	-0.0134	-0.0129	-0.0013	-0.0059	-0.0055	-0.0138
5000	+0.0114	+0.0100	+0.0103	+0.0082	+0.0008	+0.0026
10000	+0.0169	+0.0149	+0.0144	+0.0127	+0.0092	+0.0115

At $N = 100$ the ranking is XGBoost 0.7802, random forest 0.7695, QSVm-Z 0.7613, SVM-RBF 0.7296, linear 0.7261, QSVm-ZZ 0.6989: the quantum kernel is next to last, and the gap to XGBoost, -0.0812 , is the largest in the sweep. At $N = 10^4$ the ranking is QSVm-ZZ 0.7855, QSVm-Z 0.7787, SVM-RBF 0.7740, random forest 0.7728, XGBoost 0.7706: the quantum kernel is first. The paired difference

against XGBoost passes through zero between $N = 2000$ and $N = 5000$ ($-0.0129 \rightarrow +0.0100$), and is corrected-significant at both $N = 5000$ ($p_{\text{Holm}} = 0.027$, $d_z = 1.22$) and $N = 10^4$ ($+0.0149$, $p_{\text{Holm}} = 0.0078$, $d_z = 1.07$).

Three qualifications keep this from being read as more than it is.

First, the reversal is not an artefact of re-tuning. Table II repeats the sweep with hyper-parameters frozen at their C2 values instead of re-tuned at each N ; all six baseline-arm combinations change sign in the same interval, $2000 \rightarrow 5000$. A re-tuning artefact would not survive freezing the tuning.

Second, the reversal is not uniform across baselines. Against SVM-RBF the difference does turn positive, but no cell in the sweep reaches corrected significance — the quantum kernel never demonstrably beats an RBF kernel at any size we tested. Significance is reached against XGBoost at $N \geq 5000$, against the random forest only at $N = 10^4$, and against the linear and poly-2 kernels from $N = 1000$ and $N = 2000$ respectively.

Third, the reversal is a property of the natural prior, not of the data volume alone. Repeating the sweep with rare attacks enriched twelvefold to 10% — the sampling the submitted protocol used — leaves 26 of 30 comparisons inconclusive, with no crossover anywhere in range. That regime cannot be pushed past $N = 2000$: at $N = 5000$ the ten runs would

Under class-prior shift the quantum kernel degrades more than the trees

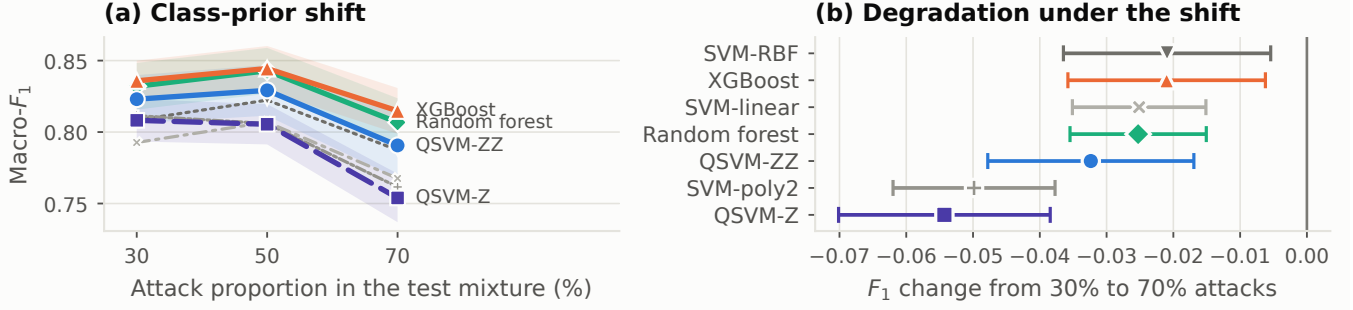


Fig. 5. **Under class-prior shift the quantum kernel degrades more than the tree ensembles, not less.** (a) Macro- F_1 as the attack proportion moves from 30% to 70%, over ten runs with 95% bands. (b) The paired change across that shift, sorted. QSVM-ZZ loses 0.032 against 0.021 for XGBoost and 0.025 for the random forest. The submitted version named this regime as its strongest evidence.

Enriching rare attacks removes the crossover: NSL-KDD

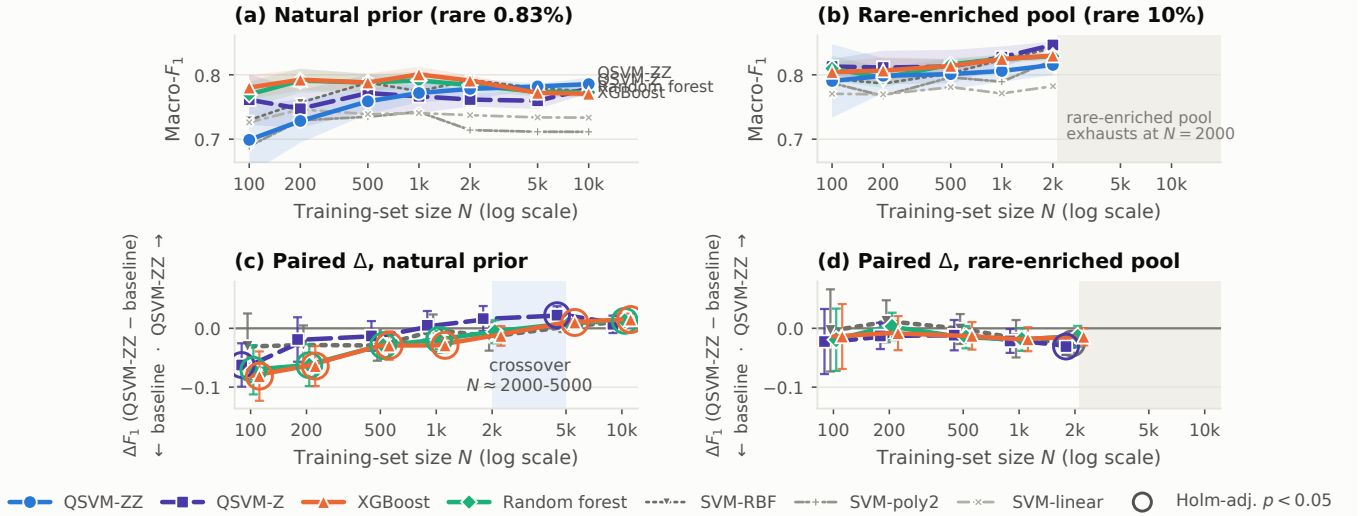


Fig. 6. **Rare-attack enrichment, not sample size alone, decides whether the quantum kernel leads on NSL-KDD.** (a, b) Mean macro- F_1 over ten runs on the full KDDTest+ set with 95% bands; (c, d) paired differences against each baseline, circled where the Holm-adjusted $p < 0.05$. Under the natural prior the ordering reverses between $N = 2000$ and $N = 5000$ (band in c). Twelvefold rare enrichment removes the reversal (d).

share 59% of their rare samples, and independence between runs would be lost. Within the range where the two regimes overlap, changing only the class prior removes the effect. This is a direct consequence for the submitted protocol, whose enrichment concealed the very regime in which the kernel does best.

E. Rare attacks

Because R2L and U2R are the classes an operator cares about, we report them separately rather than only inside the macro average. At $N = 10^4$ under the natural prior the rare-class F_1 is SVM-RBF 0.585, QSVM-zz 0.507, QSVM-Z 0.421, XGBoost 0.334, random forest 0.331, poly-2 0.109, linear 0.089.

Two things follow. The quantum kernel is far better on rare attacks than either tree ensemble, which is not visible in the macro average and is worth knowing. But SVM-RBF is better still, by 0.078, so the honest statement is that an RBF kernel is the strongest rare-attack detector in this study. The submitted version reported a +6.7-point rare-attack gain with $d = +0.68$; no rare-class metric was computed by the code that produced it, and we withdraw the number.

F. Transfer to UNSW-NB15

The selection rule is applied to UNSW-NB15 without changing a threshold or a constant (Fig. 1(b)). It returns $n^* = 6$, at $V = 0.9044$, KTA = 0.1986 and 60 two-qubit gates. Re-running it on ten independent subsets returns $n^* = 6$ in 10/10 at tolerance $\varepsilon \in \{0.02, 0.05\}$ and 7/10 at $\varepsilon = 0.1$.

On UNSW-NB15 the quantum kernel beats classical kernels, not tree ensembles

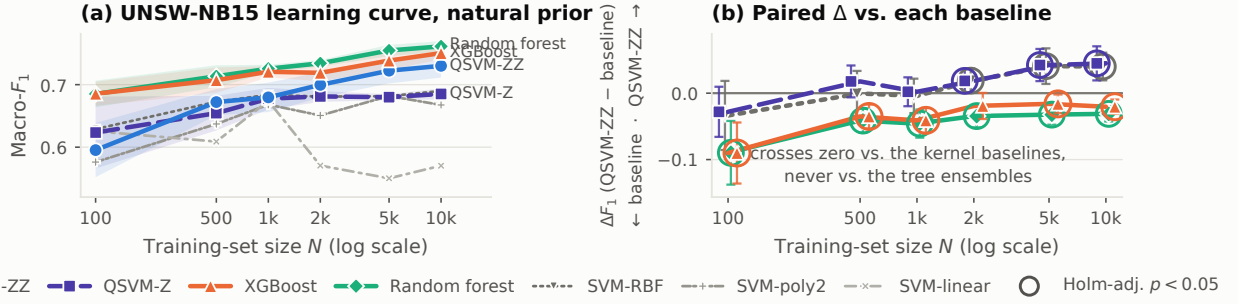


Fig. 7. **What transfers to UNSW-NB15, and what does not.** (a) Mean macro- F_1 over ten runs on the official test set; (b) paired differences, circled where the Holm-adjusted $p < 0.05$. The selection rule and the entanglement ablation transfer; the sample-complexity crossover does not. Against both tree ensembles the difference keeps its sign at every N , the opposite of NSL-KDD.

That the rule returns a different width on a different dataset is the point: it is a rule with inputs, not a constant dressed as one.

The sample-complexity result does not transfer (Fig. 7). At $N = 10^4$, QSVN-ZZ is favourable against SVM-RBF (+0.0401, $p_{\text{Holm}} = 0.0059$) and against its own entanglement-free control (+0.0449, $p_{\text{Holm}} = 0.0020$), but XGBoost stays ahead at every size tested (-0.0204 at $N = 10^4$, classical-favourable), and no crossover against either tree ensemble occurs anywhere in the sweep. A finding that reproduces on one dataset and not the other is reported as exactly that.

G. Where the kernel breaks, and why

Sec. V-A showed that a larger feature budget forces a wider circuit. Paying that price is the sharpest test of whether the quantum kernel has headroom, and it fails it (Fig. 8).

Re-running the entire sample-complexity sweep at $K = 80$, where the rule returns $n^* = 8$, produces 48 paired comparisons — six baselines across eight training-set sizes — and every one of them is classical-favourable. The entanglement ablation reverses sign as well: at eight qubits QSVN-ZZ falls *below* its entanglement-free control at every size, by 0.082 to 0.171 macro- F_1 . Widening the circuit does not extend the advantage; it removes it.

The mechanism is measurable in the Gram matrix rather than inferred. Writing $s(n)$ for the standard deviation of the off-diagonal entries and fitting $s(n) \propto n^{-\alpha}$ over $n = 4, \dots, 10$, we obtain $\alpha_{\text{ZZ}} = 0.590$ against $\alpha_{\text{Z}} = 0.292$ at $K = 20$ — a ratio of 2.02. That ratio is what the circuit structure predicts: the ZZ map contributes $\binom{n}{2}$ pairwise phase terms where the Z map contributes n single-qubit terms, so the former's phase content grows quadratically and the latter's linearly. Over the same range at $K = 80$ the ZZ kernel loses 49% of its off-diagonal dispersion against 18% for the Z kernel.

The concentration is not merely present; it tracks the end-task result. Across $n = 4, \dots, 10$ the correlation between off-diagonal dispersion and macro- F_1 is $r = +0.77$ for the ZZ kernel and $r = +0.32$ for the Z kernel, which barely

concentrates. As the Gram matrix flattens, accuracy follows it down. This is the exponential-concentration phenomenon of [22] in a concrete applied setting, and it is why we present a bounded operating window rather than a method that scales.

An independent measurement supports the boundary and sharpens its interpretation. Cirillo *et al.* [36], sweeping qubit count on the same corpus under backend-derived noise, locate good performance in a 20–35 two-qubit-gate band with degradation beyond roughly 60 gates. Our operating point of 24 gates sits inside that band and our collapse at 112 gates lies beyond their threshold, so the two studies agree on where the usable window ends. They differ on why: that study attributes the degradation to two-qubit gate error accumulating on hardware, whereas the results above are obtained under exact, noiseless simulation, in which no gate error exists. The window therefore does not widen with better hardware. It is set by the kernel.

VI. REGIME MAP: WHEN TO USE QSVN-ZZ

Collecting every controlled comparison in one place gives the map in Fig. 9: 110 cells, each a paired comparison between QSVN-ZZ and one baseline under one condition, each carrying a Holm-corrected verdict. Twenty-one favour the quantum kernel, twenty-one favour a classical baseline, and sixty-eight are inconclusive.

The balance of the first two numbers is the honest summary of this study. A practitioner reading only the count would conclude that the quantum kernel is neither better nor worse on average, and that conclusion would be correct and useless. The map is worth more than its margin because the verdicts are not distributed at random across conditions.

A. Where it is worth the cost

Three conditions produce a favourable verdict, and they share a structure.

Enough data, natural prior. At $N \geq 5000$ under the natural class prior, QSVN-ZZ is favourable against XGBoost and, at $N = 10^4$, against the random forest; it is favourable against the linear and poly-2 kernels from $N = 1000$ and $N = 2000$.

Widening the circuit destroys the quantum kernel, and the kernel geometry says why

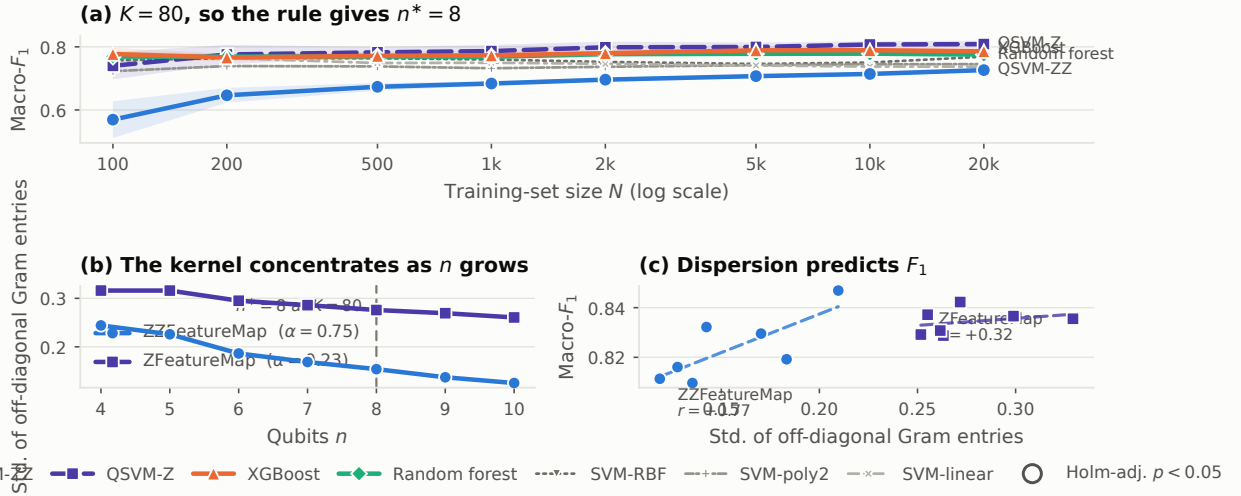


Fig. 8. **Widening the circuit destroys the quantum kernel, and the kernel geometry says why.** (a) The sample-complexity sweep re-run at $K = 80$, where the rule returns $n^* = 8$: all 48 paired comparisons are classical-favourable. (b) Off-diagonal Gram dispersion against width, fitted as $n^{-\alpha}$; the ZZ map concentrates about twice as fast as the Z map. (c) That dispersion predicts macro- F_1 for ZZ ($r = +0.77$) but not for Z ($r = +0.32$).

Regime map: where the quantum kernel helps, and where it does not

Verdict of QSVm-ZZ against each baseline (paired over 10 runs, Holm-corrected within family)

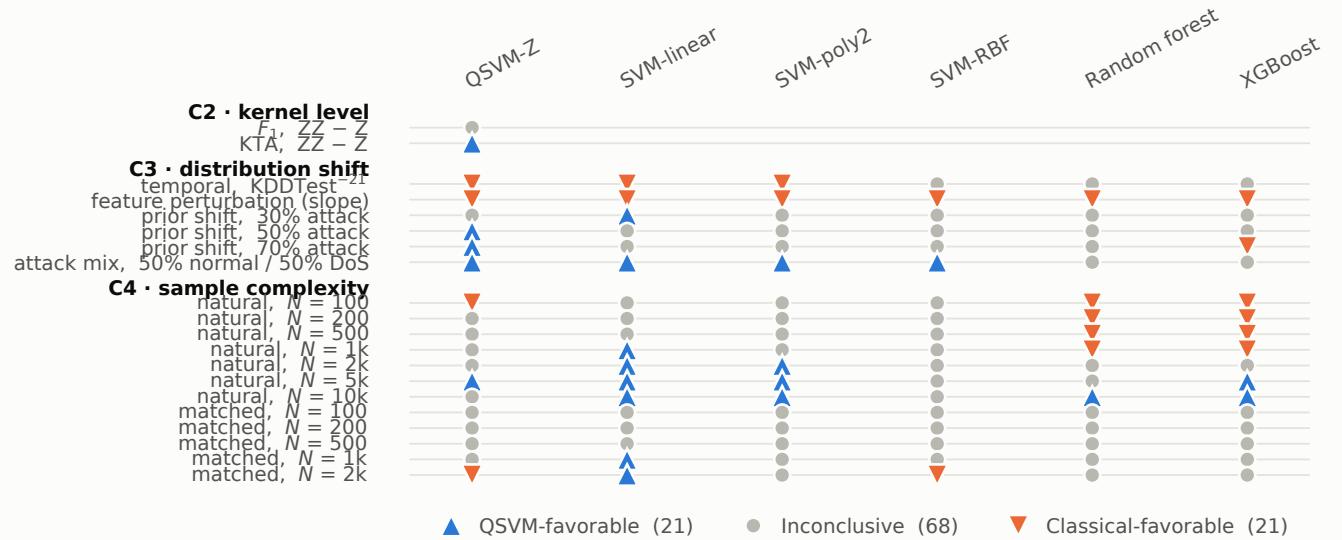


Fig. 9. **Regime map: 110 controlled comparisons of QSVm-ZZ against six baselines.** Each cell is one paired comparison over ten runs, Holm-corrected within its baseline family. Verdicts are encoded by both colour and glyph so the figure survives greyscale printing. The distribution is the result: 21 favourable, 21 classical-favourable, 68 inconclusive. A single aggregate number would hide exactly the structure this paper is about.

Below $N = 2000$ the verdicts run the other way. The regime is bounded on both sides: below it the kernel has too little data, and Sec. V-G shows that a wider circuit does not extend it upward.

Balanced binary composition. On a 50/50 Normal-DoS mixture the quantum kernel is favourable against all four kernel baselines, with effect sizes up to 3.02. This is the

condition under which quantum-kernel IDS results are most often reported in the literature, and our numbers reproduce that impression — which is part of why we report the other conditions too.

Rare-attack detection relative to tree ensembles. At $N = 10^4$ the rare-class F_1 of QSVm-ZZ is 0.507 against 0.334 for XGBoost and 0.331 for the random forest. If the deployment

objective is rare-attack recall rather than macro-averaged accuracy, the ranking changes substantially — though SVM-RBF, at 0.585, remains ahead of both.

B. Where it is not

Small training sets. At $N \leq 500$ the quantum kernel loses to both tree ensembles at every size, by up to 0.081 macro- F_1 . No configuration we tested recovers this.

Feature perturbation. Six of six comparisons are classical-favourable with effect sizes beyond 2.8. The quantum kernel is the most perturbation-sensitive model in the study, which matters for a security application where adversarial feature manipulation is the norm rather than the exception.

Strong prior shift. At a 70% attack rate the comparison against XGBoost is classical-favourable, and QSVM-ZZ degrades across the shift by 0.032 against 0.021 for XGBoost. The submitted version presented this regime as its strongest evidence; against strong tabular baselines it is evidence in the opposite direction.

Wide circuits. At $K = 80$ and $n^* = 8$, all 48 comparisons are classical-favourable. This is the sharpest boundary we found and the least expected: it says the operating window is closed from above by the physics of the feature map rather than by engineering limits that better hardware would lift.

Any comparison against SVM-RBF on sample size. Across the whole sweep, no cell reaches corrected significance in the quantum kernel's favour. An RBF kernel is the baseline this method does not clear.

C. Reading the map

Two properties of the map should govern how it is generalised.

First, the inconclusive cells are the majority, and they are informative. With ten runs and Holm correction the smallest difference that reaches a verdict anywhere in the map is 0.010 macro- F_1 , while differences as large as 0.031 remain inconclusive when their run-to-run spread is wide; the median inconclusive cell sits at 0.014. Most of the differences in this study are of that order. A study reporting sharper verdicts on the same data with fewer repetitions and no multiplicity control is not seeing more; it is resolving less.

Second, the favourable regime is bounded on every axis we varied, and the bounds are not far apart: $N \gtrsim 5000$ but not at a width beyond four to six qubits, under the natural prior but not under enrichment, on NSL-KDD but not on UNSW-NB15 for the sample-size effect. A method whose advantage is confined to a box this small is not yet a practical instrument, and we do not present it as one. What it is, on this evidence, is a phenomenon with a measurable boundary — and the boundary, being predicted by the circuit's phase-term count, is the part most likely to hold beyond these two datasets.

VII. LIMITATIONS AND THREATS TO VALIDITY

A. Hardware noise: what was and was not measured

The submitted version reported only finite-shot sampling error and deferred coherent noise to a companion study.

Reviewers correctly objected that shot noise alone cannot support a NISQ-feasibility claim. We therefore state the scope precisely.

What we measure. The C2 pipeline is re-run under a noise model derived from the calibration data of a real superconducting backend (FakeManilaV2) via `NoiseModel.from_backend`, which carries gate errors, readout errors, and gate-duration-dependent thermal relaxation. Circuits are transpiled onto that backend's coupling map and basis gate set, giving a depth of 59 and 44 two-qubit gates for the $n = 4$, $r = 2$ map. Under this model the entanglement conclusion is unchanged: the alignment gap between QSVM-ZZ and QSVM-Z survives, while the macro- F_1 difference remains within its confidence interval.

What we do not measure. The circuits were not executed on a quantum processor. Two effects are therefore outside our evidence: drift of calibration parameters between the snapshot and execution time, and idle-time decoherence beyond what gate durations already attribute. Our released code contains the execution path for a real backend, and the comparison it would produce (exact statevector $\rightarrow$ backend-derived noise $\rightarrow$ hardware) is specified, but we report no hardware numbers and make no claim about them. Readers should read “NISQ-aware” as a statement about *circuit budget* — qubit count, depth, and two-qubit gate count chosen under an explicit cost model — and not as a claim of hardware validation.

B. Embedding size: now measured, and the news is not good for width

The submitted version listed the effect of moving along the frontier as an open question. We have since measured it, and the answer constrains the method more sharply than we expected.

Increasing the feature budget K does buy accuracy for a classical proxy, but it is not free: retaining 85% of the variance at $K = 80$ requires $n^* = 8$ qubits rather than the $n^* = 4$ that $K = 20$ admits, raising the two-qubit gate count from 24 to 112. Re-running the full sample-complexity sweep at that operating point removes the quantum kernel's advantage entirely: across eight training-set sizes and six baselines, all 48 paired comparisons favour the classical model, and the entanglement ablation reverses sign, with QSVM-ZZ now below its entanglement-free control.

The mechanism is measurable rather than conjectural. The dispersion of the off-diagonal Gram entries decays with width as $n^{-\alpha}$, and the fitted exponent for the ZZ map is close to twice that of the Z map at $K = 20$ (0.59 vs. 0.29), consistent with the $\binom{n}{2}$ pairwise phase terms of the former growing quadratically where the latter's grow linearly. Across $n = 4$ to 10 the macro- F_1 of the ZZ kernel tracks this dispersion ($r = +0.77$), whereas the Z kernel, which barely concentrates, shows no such dependence ($r = +0.32$). This is the concentration phenomenon of [22] appearing directly in our setting.

Two caveats. The exponent ratio is close to 2 at $K = 20$ but larger at $K = 80$ (3.32), so the term-counting argument predicts the ordering but not the full rate. And the F_1 -dispersion relation is measured on seven widths at a single

training-set size; we present it as an empirical regularity, not a law.

C. Classical-baseline strength

The submitted version compared only against SVM variants. We have added a random forest and XGBoost under the identical protocol, and they change the headline: on the main benchmark XGBoost attains the highest mean macro- F_1 (0.8503) ahead of QSVM-ZZ (0.8469), with overlapping intervals. We report this as an ordering of point estimates, not a significant separation.

We did not add CatBoost or deep tabular models such as TabNet and FT-Transformer. CatBoost would supply a second gradient-boosting representative without changing the comparison structure; the deep tabular family would require its own tuning-budget protocol to be compared fairly, which would enlarge the study from a controlled quantum-kernel benchmark into a survey of tabular architectures. We therefore do not claim coverage of all strong tabular methods, and any practical reading of our results should be qualified accordingly.

D. Dataset breadth and protocol

The submitted version rested almost entirely on NSL-KDD. We have added UNSW-NB15 under the same protocol, which is what allows us to separate findings that transfer from findings that do not: the dimension-selection rule and the entanglement ablation transfer, while the sample-complexity crossover and the rare-class behaviour do not.

Two properties of the data limit what either dataset can show. NSL-KDD is an aging benchmark in which 610 test rows are identical to a training row in both features and label (617 match on features alone). UNSW-NB15 is far more heavily duplicated: 47.3% of training rows are internal repeats, and 25.0% of test rows have an exact duplicate in the training partition, which inflates absolute scores for every model alike. Because our claims are paired differences between models evaluated on the same rows, this affects the level rather than the comparison, but it does mean the absolute numbers here are not comparable to studies that de-duplicate.

Finally, the feature budget $K = 20$ is inherited from the submitted protocol and held fixed so that the revised results remain comparable to it. Fig. 2 shows it is not the accuracy optimum; it is the value at which the selection rule returns a four-qubit circuit.

VIII. CONCLUSION AND FUTURE WORK

We set out to locate, rather than to demonstrate, the value of a NISQ-feasible quantum kernel for network intrusion detection. Varying one factor at a time over 110 controlled comparisons, we find 21 verdicts favouring the quantum kernel, 21 favouring a classical baseline, and 68 inconclusive — and a structure inside that balance that a single-operating-point benchmark cannot see.

The structure has three parts. The ordering between QSVM-ZZ and every strong classical baseline reverses between $N = 2000$ and $N = 5000$ under the natural class prior, in both

tuning arms, and disappears when rare attacks are enriched. A lexicographic selection rule that charges for two-qubit gates returns $n^* = 4$ on NSL-KDD and $n^* = 6$ on UNSW-NB15 without any change of threshold, so the embedding dimension is a property of the dataset rather than a constant to be inherited. And the operating window is closed from above: at $K = 80$, where the rule requires eight qubits, every one of 48 comparisons favours the classical model, with off-diagonal Gram dispersion decaying about twice as fast for the ZZ map as for its entanglement-free control and tracking macro- F_1 at $r = +0.77$.

That last result is the one we would most like others to test. It ties a structural property of the circuit — $\binom{n}{2}$ pairwise phase terms against n single-qubit ones — to a decay exponent we can measure and to an end-task metric that follows it down. If it holds on other corpora and other maps, then the practical question for quantum kernels on tabular data is not how to make them more expressive but how to keep them from concentrating, and the useful width is bounded well below where hardware limits bite.

We are equally clear about what this study does not establish. The circuits were not executed on a quantum processor, and “NISQ-aware” here means a circuit budget chosen under an explicit cost model, not hardware validation. The comparison covers two datasets, two tree ensembles and three kernel baselines, not the deep tabular architectures. The favourable regime is narrow on every axis we varied. And five claims from the earlier version of this work do not survive the revised protocol; they are withdrawn in Sec. I and corrected in Sec. III-H.

Three directions follow directly. Executing the same kernels on hardware would close the one gap our simulations cannot: the released code contains the execution path and the three-tier comparison it would produce is already specified. Testing the dispersion–accuracy relationship on projected quantum kernels, which concentrate differently, would show whether the boundary we measured is a property of fidelity kernels specifically or of the approach. And since the crossover reproduces on one dataset and not the other, the mechanism behind it — why more data helps this kernel more than it helps a tree ensemble, and only under a skewed prior — remains open. Our full implementation, protocol files, intermediate artifacts and two independent audit scripts are released so that each of these can be started from where we stopped.

APPENDIX A PROOF OF LEMMA 1

Let $r = 1$, write $|+\rangle := H^{\otimes n}|0\rangle^{\otimes n}$, and let $\Delta := x - x'$. From Eq. (1),

$$K^Q(x, x') = |\langle + | U_\Phi(x')^\dagger U_\Phi(x) | + \rangle|^2.$$

Both factors are diagonal in the computational basis, so their product is the diagonal unitary $\text{diag}(e^{i\theta(s)})$ indexed by the sign vectors $s \in \{-1, +1\}^n$ that label the Z -eigenbasis, with, from Eq. (2),

$$\theta(s) = \sum_{i=1}^n \Delta_i s_i + \frac{1}{2} \sum_{1 \leq i < j \leq n} \Delta \varphi_{ij} s_i s_j. \quad (9)$$

Because $|+\rangle$ places equal amplitude $2^{-n/2}$ on every basis state,

$$\langle + | U_{\Phi}(\mathbf{x}')^{\dagger} U_{\Phi}(\mathbf{x}) | + \rangle = \mathbb{E}_{\mathbf{s}}[e^{i\theta(\mathbf{s})}], \quad (10)$$

the expectation being over $\mathbf{s}$ uniform on $\{-1, +1\}^n$.

Under that distribution the parity functions s_i and $s_i s_j$ are orthonormal: each has mean zero and unit variance, and distinct ones are uncorrelated. Hence $\mathbb{E}[\theta] = 0$ and, by Eq. (9),

$$\text{Var}(\theta) = \mathbb{E}[\theta^2] = \sum_{i=1}^n \Delta_i^2 + \frac{1}{4} \sum_{1 \leq i < j \leq n} \Delta \varphi_{ij}^2. \quad (11)$$

Expanding Eq. (10) and using $\mathbb{E}[\theta] = 0$,

$$\mathbb{E}[e^{i\theta}] = 1 - \frac{1}{2} \mathbb{E}[\theta^2] - \frac{i}{6} \mathbb{E}[\theta^3] + O(\|\Delta\|^4),$$

so that, writing $a = \frac{1}{2} \mathbb{E}[\theta^2]$ and collecting the purely imaginary remainder into $b = O(\|\Delta\|^3)$,

$$|\mathbb{E}[e^{i\theta}]|^2 = (1 - a)^2 + b^2 = 1 - \mathbb{E}[\theta^2] + O(\|\Delta\|^4).$$

Substituting Eq. (11) gives Eq. (5). Note that the cubic term contributes only at fourth order after squaring, which is why the remainder is $O(\|\Delta\|^4)$ and not $O(\|\Delta\|^3)$. ■

The step that fails for $r \geq 2$ is the first: with r repetitions the Hadamard layers separate the diagonal factors, the product is no longer diagonal, and Eq. (10) does not apply (Remark 2). Against the exact statevector kernel, itself checked against Qiskit to within 1.3×10^{-15} , a least-squares fit of $1 - K^Q = a \sum_i \Delta_i^2 + b \sum_{i < j} \Delta \varphi_{ij}^2$ at $r = 1$ recovers $a = 1.000$ and $b = 0.250$ with $R^2 = 1.000000$ at $n = 3, 4, 6$, and the residual of Eq. (5) decays as $\|\Delta\|^{3.9}$, reaching 5×10^{-11} at $\|\Delta\| = 10^{-3}$. At $r = 2$ the same fit attains only $R^2 = 0.60$ to 0.94 while the unrestricted quadratic form still attains $R^2 = 0.999996$. The script is released as `runners/verify_lemmal.py`.

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